

Hashing Patterns for DSA

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1 Introduction

Hashing stores key-value pairs for average $O(1)$ lookup, insertion, and deletion.

2 STL

2.1 unordered_map

Used for frequencies, prefix sums, and lookups.

```
1 mp[x]++;  
2 if (mp.count(x)) {}  
3 mp.erase(x);
```

2.2 unordered_set

Used when only existence matters.

```
1 st.insert(x);  
2 st.count(x);
```

3 Complexities

Container	Search	Insert	Delete
unordered_map	$O(1)$ avg	$O(1)$ avg	$O(1)$ avg
unordered_set	$O(1)$ avg	$O(1)$ avg	$O(1)$ avg
map	$O(\log n)$	$O(\log n)$	$O(\log n)$
set	$O(\log n)$	$O(\log n)$	$O(\log n)$

4 Core Patterns

4.1 Frequency Counting

Questions:

- Majority Element
- Top K Frequent Elements
- Valid Anagram
- Group Anagrams

Template:

```
1 for(int x: nums) freq[x]++;
```

4.2 Lookup Pattern

Typical problem: Two Sum.

```
1 for(int i=0;i<nums.size();i++){
2     if(mp.count(target-nums[i]))
3         return {mp[target-nums[i]],i};
4     mp[nums[i]]=i;
5 }
```

4.3 Prefix Sum + Hash Map

Important problems:

- Subarray Sum Equals K
- Binary Subarrays With Sum
- Contiguous Array

Template:

```
1 unordered_map<int,int> mp;
2 mp[0]=1;
3 int sum=0,ans=0;
4 for(int x:nums){
5     sum+=x;
6     ans+=mp[sum-k];
7     mp[sum]++;
8 }
```

4.4 Prefix XOR

Use XOR instead of sum.

```
1 mp[0]=1;
```

4.5 Sliding Window + Hashing

Problems:

- Longest Substring Without Repeating Characters
- Minimum Window Substring
- Find All Anagrams

5 Recognition Table

Question Clue	Pattern
Count occurrences	Frequency map
Pair equals target	Lookup map

Count subarrays	Prefix sum + map
XOR subarrays	Prefix XOR
Longest consecutive	Hash set
Distinct window	Sliding window + map
Modulo condition	Prefix modulo

6 Common Mistakes

- Forgetting `mp[0]=1` for prefix sum counting.
- Overwriting first occurrence when longest subarray is required.
- Using `map` instead of `unordered_map` unnecessarily.

7 Important Interview Problems

1. Two Sum
2. Contains Duplicate
3. Majority Element
4. Valid Anagram
5. Group Anagrams
6. Longest Consecutive Sequence
7. Subarray Sum Equals K
8. Binary Subarrays With Sum
9. Count Nice Subarrays
10. Contiguous Array
11. Continuous Subarray Sum
12. Subarray XOR Equals K
13. Longest Substring Without Repeating Characters
14. Minimum Window Substring
15. Find All Anagrams in a String